

DISTRIBUTIONAL DYNAMICS

Christian Bayer

*University of Bonn,
CEPR, IZA, and CESifo*

Luis Calderon

University of Bonn

Moritz Kuhn

*University of Mannheim,
CEPR, and IZA*

May 10, 2026



MOTIVATION

The last decade has seen an explosion of work on the **importance of household heterogeneity** for macroeconomic dynamics.

MOTIVATION

The last decade has seen an explosion of work on the **importance of household heterogeneity** for macroeconomic dynamics.

However there is limited availability of data on household heterogeneity that:

- ▶ is of high-frequency
- ▶ and **links** consumption, income, and wealth.

WHAT DATA IS CURRENTLY OUT THERE?

Commonly used micro data on U.S. household heterogeneity:

PSID

CEX

SCF

CPS

SIPP

WHAT DATA IS CURRENTLY OUT THERE?

Commonly used micro data on U.S. household heterogeneity:

- ▶ **may not have all variables of interest covered**

e.g., which datasets contain consumption?

PSID

CEX

SCF

CPS

SIPP

WHAT DATA IS CURRENTLY OUT THERE?

Commonly used micro data on U.S. household heterogeneity:

- ▶ **may not have all variables of interest covered**

e.g., contains consumption and wealth?

PSID

CEX

SCF

CPS

SIPP

WHAT DATA IS CURRENTLY OUT THERE?

Commonly used micro data on U.S. household heterogeneity:

- ▶ may not have all variables of interest covered
- ▶ **is of mixed frequency**

PSID

CEX

SCF

CPS

SIPP

WHAT DATA IS CURRENTLY OUT THERE?

Commonly used micro data on U.S. household heterogeneity:

- ▶ may not have all variables of interest covered
- ▶ **is of mixed frequency**

e.g., data observed every quarter

PSID

CEX

SCF

CPS

SIPP

WHAT DATA IS CURRENTLY OUT THERE?

Commonly used micro data on U.S. household heterogeneity:

- ▶ may not have all variables of interest covered
- ▶ **is of mixed frequency**

e.g., data observed every year

PSID

CEX

SCF

CPS

SIPP

WHAT DATA IS CURRENTLY OUT THERE?

Commonly used micro data on U.S. household heterogeneity:

- ▶ may not have all variables of interest covered
- ▶ **is of mixed frequency**

e.g., consistent sampling intervals

PSID

CEX

SCF

CPS

SIPP

WHAT DATA IS CURRENTLY OUT THERE?

Commonly used micro data on U.S. household heterogeneity:

- ▶ may not have all variables of interest covered
- ▶ is of mixed frequency
- ▶ **carry differences in operationalizations and noise**

*e.g., different income concepts, sampling procedures,
sample sizes, etc*

PSID

CEX

SCF

CPS

SIPP

CONTRIBUTION AND PREVIEW OF RESULTS

Our contribution allows us to generate consensus estimates of joint distributions e.g., income, wealth and consumption

- ▶ ... at a **balanced, high frequency**
- ▶ ... for **any granularity**
- ▶ ... for a **long time horizon**
- ▶ ... using **any set** of publicly available data

CONTRIBUTION AND PREVIEW OF RESULTS

Our contribution allows us to generate consensus estimates of joint distributions e.g., income, wealth and consumption

- ▶ ... at a **balanced, high frequency**
- ▶ ... for **any granularity**
- ▶ ... for a **long time horizon**
- ▶ ... using **any set** of publicly available data

I will later present exercises that speak to the quality of these estimates.

LITERATURE REVIEW (1/2)

We speak to several strands of literature:

▶ **Household heterogeneity for modeling macroeconomic dynamics:**

[kaplan2018monetary; Bayeretal2019; bayer2020shocks; krueger2016macroeconomics]

▶ **Household heterogeneity for MP & FP transmission:** [berger2023imperfect;

coibion2017innocent; cloyne2020monetary; holm2021transmission; chang2024effects; bartscher2022monetary]

▶ **Time series in inequality:** [piketty2003income; saez2016wealth; piketty2018distributional; batty2020distributional;

smith2021top; blanchet2022real]

▶ **Interpolation and prediction of time-series:** [harvey1984estimating; harvey1990forecasting;

harvey2000estimating; monch2005towards; moauro2005temporal; proietti2006temporal]

▶ **Functional data-analysis:** [kneip2001inference; diebold2006forecasting; chang2016nonstationarity; inoue2018effects;

otto2022approximate; chang2024heterogeneity]

LITERATURE REVIEW (2/2)

The closest paper to ours is **Chang, Chen, and Schorfheide (JPE, 2024)**.

LITERATURE REVIEW (2/2)

The closest paper to ours is **Chang, Chen, and Schorfheide (JPE, 2024)**.

	Their paper	Our Paper
Focus	Joint dynamics of the cross-sectional earnings density	Production of high quality, synthetic data
Contribution		
Multi-dimensionality	Data Utilization	Handling of Data
Estimates joint distributions	Able to use all publicly available data	Handles missing and mixed frequency data both over time and scope

Methodology

MODELING THE DYNAMICS

Let $\Xi_t(w)$ be some **joint distribution** over a random vector w and Y_t be aggregate variables including lags. To model the **dynamics**, we assume the density $d\Xi_t(w)$ evolves **according to the difference equation**:

$$d\Xi_t(w) = G(d\Xi_{t-1}(w), Y_t) + \varepsilon_t(w)$$

- where $\varepsilon_t(w)$ is a random element in the $\mathcal{L}^2$ functional space.

ESTIMATING THE DYNAMICS: DIFFICULTIES TO OVERCOME

Estimating $d\Xi_t(w)$, **even with finite support**, has the following difficulties:

$$d\Xi_t(w) = G(d\Xi_{t-1}(w), Y_t) + \varepsilon_t(w)$$

ESTIMATING THE DYNAMICS: DIFFICULTIES TO OVERCOME

Estimating $d\Xi_t(w)$, **even with finite support**, has the following difficulties:

$$d\Xi_t(w) = G(d\Xi_{t-1}(w), Y_t) + \varepsilon_t(w)$$

- (1) The density $d\Xi_t(w)$ has **time-varying support**;

ESTIMATING THE DYNAMICS: DIFFICULTIES TO OVERCOME

Estimating $d\Xi_t(w)$, **even with finite support**, has the following difficulties:

$$d\Xi_t(w) = G(d\Xi_{t-1}(w), Y_t) + \varepsilon_t(w)$$

- (1) The density $d\Xi_t(w)$ has **time-varying support**;
- (2) w is **multi-dimensional, observed** $\forall t$ and at times, **incompletely**: $\mathbf{w}_t^j \subset \mathbf{w}$;

ESTIMATING THE DYNAMICS: DIFFICULTIES TO OVERCOME

Estimating $d\Xi_t(w)$, **even with finite support**, has the following difficulties:

$$d\Xi_t(w) = G(d\Xi_{t-1}(w), Y_t) + \varepsilon_t(w)$$

- (1) The density $d\Xi_t(w)$ has **time-varying support**;
- (2) w is **multi-dimensional**, **observed** $\forall t$ and at times, **incompletely**: $\mathbf{w}_t^j \subset \mathbf{w}$;
- (3) rendering the **high-dimensional functional** difference equation feasible

(1) TIME-VARYING SUPPORT AND FUNCTIONAL INTERPRETATION

We decompose the joint distribution into two objects:

$$dC_t(u_1, \dots, u_m, \dots, u_d) \quad \Xi_{mt}^{-1}(u_m) \quad \text{for } u_m \sim U[0, 1] \quad \forall t \in \mathbb{T}, m \in \mathbb{D}$$

- ▶ dC_t the copula density;
- ▶ Ξ_{mt}^{-1} its quantile functions

How do we estimate dC_t and Ξ_{mt}^{-1} ?

(1) TIME-VARYING SUPPORT AND FUNCTIONAL INTERPRETATION

$$dC_t(u_1, \dots, u_m, \dots, u_d) = \sum_{o \in \mathbb{N}^d} \kappa_{o,t} \prod_{m=1}^d Q_{o_m}(u_m)$$

$$\Xi_{mt}^{-1}(u_m) = \sum_{o \in \mathbb{N}} \xi_{o,m,t} Q_o(u)$$

(1) TIME-VARYING SUPPORT AND FUNCTIONAL INTERPRETATION

$$dC_t(u_1, \dots, u_m, \dots, u_d) = \sum_{o \in \mathbb{N}^d} \kappa_{o,t} \prod_{m=1}^d Q_{o_m}(u_m)$$

$$\Xi_{mt}^{-1}(u_m) = \sum_{o \in \mathbb{N}} \xi_{o,m,t} Q_o(u)$$

What about the coefficients $\{\kappa, \xi\}$?

(1) TIME-VARYING SUPPORT AND FUNCTIONAL INTERPRETATION

$$dC_t(u_1, \dots, u_m, \dots, u_d) = \sum_{o \in \mathbb{N}^d} \kappa_{o,t} \prod_{m=1}^d Q_{o_m}(u_m)$$

$$\Xi_{mt}^{-1}(u_m) = \sum_{o \in \mathbb{N}} \xi_{o,m,t} Q_o(u)$$

What about the coefficients $\{\kappa, \xi\}$?

$$\kappa_{o,t} \approx N^{-1} \sum_i \prod_{m=1}^d Q_{o_m}(u_{m,i,t}) \quad \xi_{o,m,t} \approx N^{-1} \sum_i w_{m,i,t} Q_o(u_{m,i,t})$$

► where $w_{m,i,t}$ will be the raw data and $u_{m,i,t}$ the respective rank

(1) TIME-VARYING SUPPORT AND FUNCTIONAL INTERPRETATION

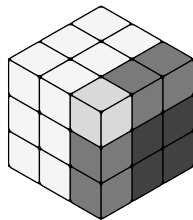
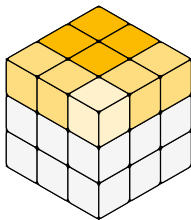
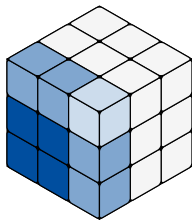
Coefficients $\{\xi, \kappa\}$ are **infinitely-many** $\implies$ truncation $\mathbb{O} \subset \mathbb{N}^d$

$$dC_t(u_1, u_2, \dots, u_d) \approx \sum_{o \in \mathbb{O}^d} \hat{\kappa}_{o,t} \prod_{m=1}^d Q_{o_m}(u_m) \quad \Xi_{mt}^{-1}(u_m) \approx \sum_{o \in \mathbb{O}} \hat{\xi}_{o,m,t} Q_o(u)$$

As a result of this joint distributional approximation:

- ▶ The support across time is always $u_m \in [0, 1]$;
- ▶ Tracking density amounts to tracking finitely many coefficients;
- ▶ Interpretation remains functional.

(2) DEALING WITH PARTIAL OBSERVABILITY OF w



- ▶ **For demonstration** let the darker edge be the leading coefficient $(1, 1, 1)$
- ▶ Each cube highlights the slice estimated for some u_d integrated out
- ▶ For example, for the **gray** slice, the third dimension is integrated out
 - ▶ copula coefficients $\kappa_{2D} = \kappa_{3D, \text{some slice}}$

▶ Proof

(3) FEASIBILITY OF ESTIMATION

Static curse of dimensionality

Since $\{\xi, \kappa\} = \hat{\phi}$ is still high-dimensional, we propose the following decomposition:

$$\hat{\phi} = \Gamma F + \gamma f$$

$$\hat{\phi} \approx \Gamma F$$

(3) FEASIBILITY OF ESTIMATION

Static curse of dimensionality

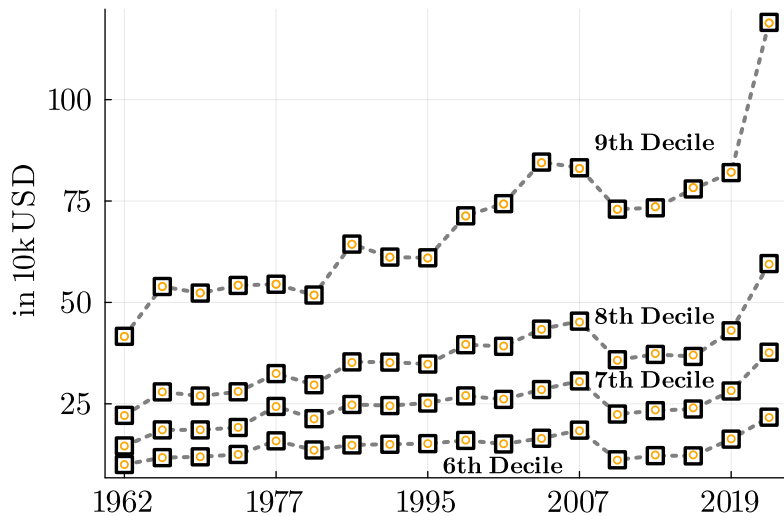
Since $\{\xi, \kappa\} = \hat{\phi}$ is still high-dimensional, we propose the following decomposition:

$$\hat{\phi} = \Gamma F + \gamma f$$

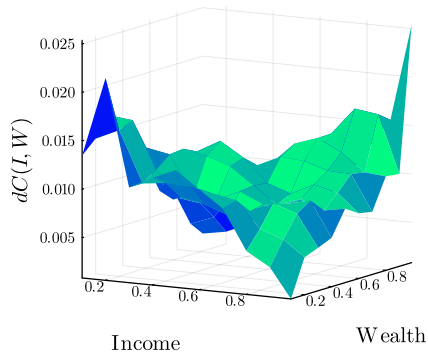
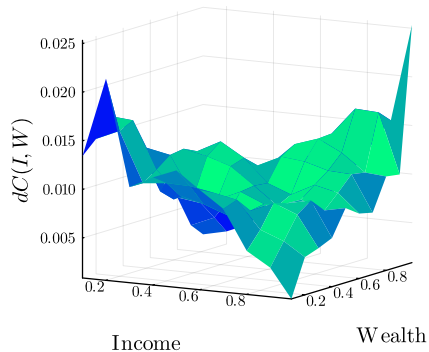
$$\hat{\phi} \approx \Gamma F$$

- ▶ where Γ is a projection matrix which maps the vector of $\hat{\phi}$ to a smaller, but ideal subspace spanned by F ;
- ▶ obtained by a singular value decomposition (SVD) after some standardization;

RECONSTRUCTIONS BETWEEN $\hat{\phi}$ AND $\approx \hat{\phi}$ LOOK NO DIFFERENT



NO DIFFERENT FOR COPULAS EITHER



► Approximation on the right

(3) FEASIBILITY OF ESTIMATION

Dynamic curse of dimensionality

The size of F will determine the size of the state-space model we will propose later ...

- ▶ ... but, we need still account for:
 - ▶ its potentially dense parameter space
 - ▶ and data sparsity

⇒ Immediately lends for a **Bayesian approach** to reduce dimensionality

Model

STATE-SPACE MODEL

We pose a linear-gaussian-state-space model to estimate the distributional dynamics:

State equation:

$$F_{t+1} = AF_t + BY_{t+1} + \epsilon_{t+1}, \quad \epsilon_t \sim \mathcal{N}(0, \Omega) \quad (1)$$

- ▶ where F_t is the vector of states, whose size is determined via the SVD;
- ▶ Y_{t+1} : aggregate factors, e.g., real variables, prices, HHs' balance sheets ...
- ▶ ϵ_{t+1} is a one-step ahead forecast Gaussian-error.

STATE-SPACE MODEL

We pose a linear-gaussian-state-space model to estimate the distributional dynamics:

State equation:

$$F_{t+1} = AF_t + BY_{t+1} + \epsilon_{t+1}, \quad \epsilon_t \sim \mathcal{N}(0, \Omega) \quad (1)$$

- ▶ we assume ϵ_{t+1} to have **diagonal** Ω ;
- ▶ A, B are unrestricted, time-invariant matrices;

STATE-SPACE MODEL

State equation w/ (unstacked) measurement equation:

$$\begin{aligned} F_{t+1} &= AF_t + BY_{t+1} + \epsilon_{t+1}, & \epsilon_t &\sim \mathcal{N}(0, \Omega) \\ \hat{\phi}_{t+1}^j &= H_{t+1}^j(\mathbf{C}F_{t+1} + v_{t+1}^j), & v_{t+1}^j &\sim \mathcal{N}(0, \mathbf{D}_j) \end{aligned} \tag{2}$$

- ▶ where $\hat{\phi}_{t+1}^j$ is a vector of coefficients;
- ▶ where H_{t+1}^j is the selector matrix which handles missing observations;

STATE-SPACE MODEL

State equation w/ (unstacked) measurement equation:

$$\begin{aligned} F_{t+1} &= AF_t + BY_{t+1} + \epsilon_{t+1}, & \epsilon_t &\sim \mathcal{N}(0, \Omega) \\ \hat{\phi}_{t+1}^j &= H_{t+1}^j(\mathbf{C}F_{t+1} + v_{t+1}^j), & v_{t+1}^j &\sim \mathcal{N}(0, \mathbf{D}_j) \end{aligned} \quad (2)$$

- ▶ $\mathbf{C}$ and $\mathbf{D}_j$ difficult to estimate jointly with A and B (numerically intense);
 - ▶ take $\mathbf{C} := \Gamma$ from the SVD;
 - ▶ let Σ_j be the sampling VCV of $\hat{\phi}^j$
 - ▶ assume $\mathbf{D}_j = \Sigma_j^{1/2} \Delta_j \Sigma_j^{1/2'}$; Δ_j is diagonal
 - ▶ i.e., D_j proportional to sampling uncertainty

STATE-SPACE MODEL

Measurement equation stacked:

$$\hat{\phi}_{t+1}^j = H_{t+1}^j(\Gamma F_{t+1} + v_{t+1}^j), \quad v_{t+1}^j \sim \mathcal{N}(0, \Sigma_j^{1/2} \Delta_j \Sigma_j^{1/2'}) \quad (3)$$

$$\mathbf{x}_{t+1} = H_{t+1} \tilde{\Gamma} F_{t+1} + \tilde{v}_{t+1}, \quad \tilde{v}_{t+1} \sim \mathcal{N}(0, \Delta) \quad (4)$$

Two steps now:

1. pre-multiplication of $\Sigma_j^{-1/2}$ for each j ;
 2. stacking all the equations together.
- Δ is diagonal and has 1 variance parameter per object-dataset pair

BAYESIAN APPROACH

Prior

$$p(\theta) = p(A, B, \Omega, \Delta) = p(A, B) \times p(\Omega) \times p(\Delta)$$

$$A, B \sim \mathcal{N}(AB_{Minn}, V_{AB_{Minn}})$$

$$\exp(\omega_{oj}) \sim \text{Lognormal}(\mu, V_{\omega})$$

$$\exp(\delta_{oj}) \sim \text{Lognormal}(\mu, V_{\delta})$$

(5)

- ▶ θ defines the parameter space;
- ▶ $AB_{Minn}, V_{AB_{Minn}}$ are initialized with a Minnesota-type prior;
- ▶ ω_{oj} and δ_{oj} are the variances of the state and measurement noise

Results

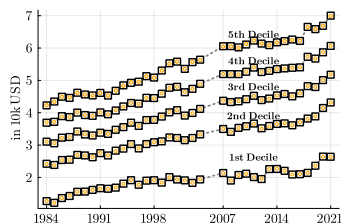
First: how reliable are the synthetic data?

Three validation exercises

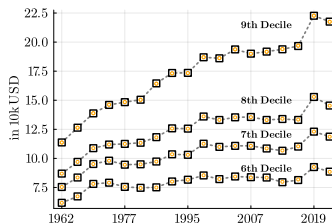
1. Factor approximation preserves the data
2. Predicting omitted survey waves
3. Recovering known dynamics from a HANK simulation

VALIDATION 1: FACTOR APPROXIMATION IS NEAR-LOSSLESS

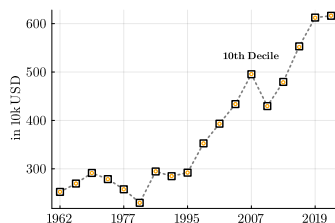
Quantile functions: raw data vs. 8-factor approximation



Consumption (bot.)



Income (mid.)

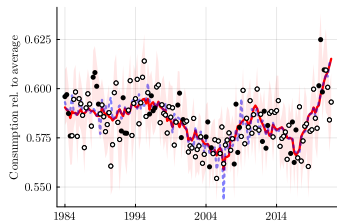


Wealth (top)

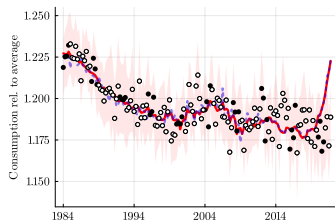
- ▶ 8 factors explain $\approx 99\%$ of business-cycle variation in $\hat{\phi}$
- ▶ KL-divergence of approximated vs. raw copula is two orders of magnitude smaller than the divergence captured by time-variation

VALIDATION 2: PREDICTING OMITTED SURVEY WAVES

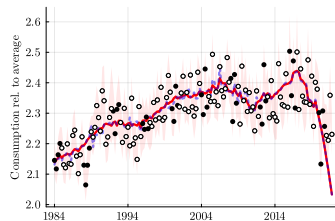
CEX every 4 years (mimicking European frequency)



Bottom 50



Next 40

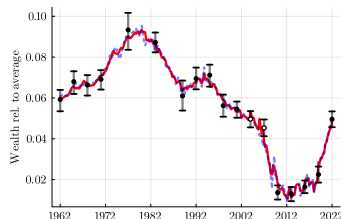


Top 10

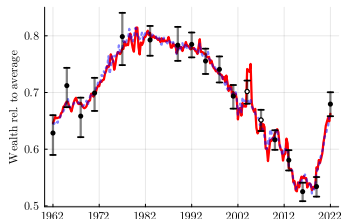
- ▶ Black dots: observations *kept*; empty dots: *dropped* from the smoother
- ▶ Solid line: baseline (all data); dashed: smoother with 1/4 of CEX waves
- ▶ Aggregate factors + other microdata $\Rightarrow$ the model fills the gaps

VALIDATION 2 (CONT.): REMOVING WEALTH ACROSS THE HOUSING CYCLE

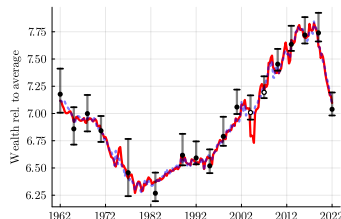
SCF wealth, with 2004Q1–2009Q4 microdata removed



Bottom 50



Next 40



Top 10

- ▶ Even *contiguous* 6-year drop in SCF wealth microdata is well-imputed
- ▶ Aggregates (house prices, balance sheets) carry the cycle through

VALIDATION 3: HANK MONTE CARLO

Setup: use the Bayer–Born–Luetticke (2024) HANK as the data-generating process.

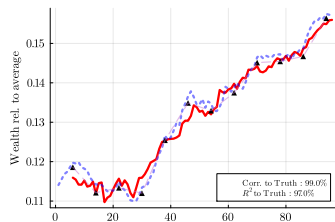
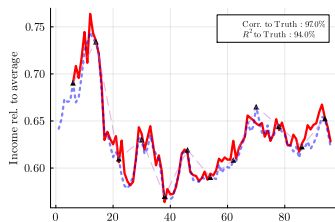
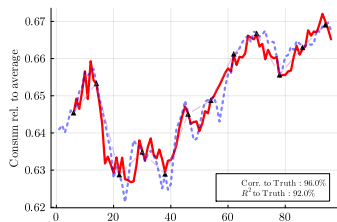
- ▶ Simulate 10 economies of 100 quarters each
- ▶ Draw 4 survey-style samples mimicking PSID / CPS / CEX / SCF
- ▶ Feed these samples to our procedure → DDFM estimates
- ▶ Compare to the *known* simulated truth

Correlation with simulated truth (avg. across economies)

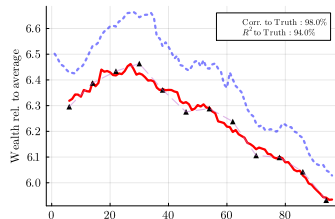
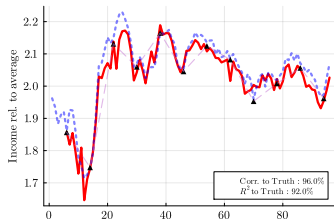
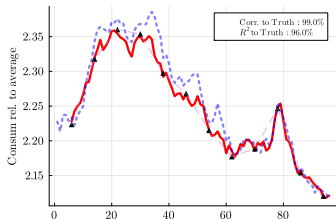
Group	<i>Consumption</i>		<i>Income</i>		<i>Wealth</i>	
	DDFM	Linear	DDFM	Linear	DDFM	Linear
Bottom 50	96	84	98	72	96	96
Middle 50–90	97	84	94	49	81	79
Top 10	93	42	81	45	87	31

VALIDATION 3 (CONT.): HANK TRUTH VS. DDFM ESTIMATES

Bottom 50 % in consumption / income / wealth (one simulated economy)



Top 10 %



So what can we do with the synthetic data?

Two applications

1. What drives distributional fluctuations?
2. Consumption dynamics across the last three U.S. recessions

APPLICATION 1: WHAT DRIVES DISTRIBUTIONAL FLUCTUATIONS?

Forecast-error variance decomposition (5-year horizon, blocked Cholesky, aggregates ordered first):

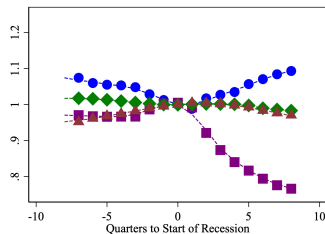
Distributional Factor No.	1	2	3	4	5	6	7	8
Aggregate shocks	99	99	70	89	93	86	86	77
Distributional shocks	1	1	30	11	7	14	14	23

- ▶ Aggregate shocks explain **at least 70%** of every factor (typically $>85\%$)
- ▶ Same story for conditional means of C , Y , W across deciles
- ▶ Echoes [bayer2020shocks]: aggregates shape distributions

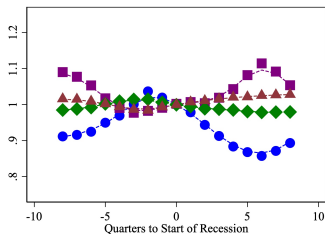
APPLICATION 2: CONSUMPTION IN THE LAST THREE RECESSIONS

Relative consumption indexed to recession peak ($t = 0$), by **joint** $Y \times W$ groups

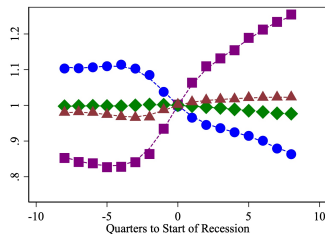
Dot-com bust



Global Financial Crisis



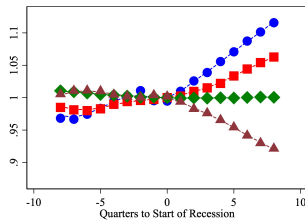
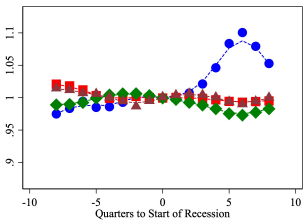
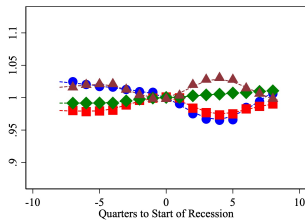
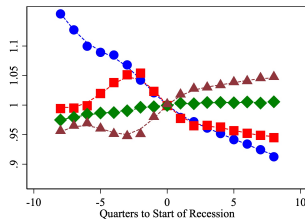
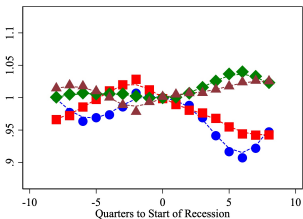
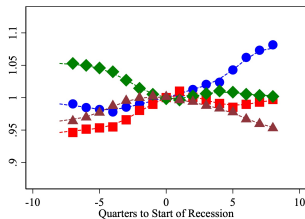
COVID-19



4 groups: **liquidity-constrained** (low Y , low W) • **asset-rich low- Y** (low Y , high W) • **middle class** • **rich** (high Y , high W)

► Distributional consequences are very different across recessions

THE SAME PICTURE marginally: INCOME VS. WEALTH

By income groups*By wealth groups*

TAKE-AWAY: STATE-DEPENDENCE IN RECESSIONS

- ▶ **Dot-com:** a *wealth* shock $\Rightarrow$ asset-rich low-income households hit hardest; low-wealth households gain relative to average
- ▶ **Global Financial Crisis:** a *balance-sheet* shock $\Rightarrow$ low-wealth, indebted households suffer sharpest consumption declines
- ▶ **COVID-19:** an *income + transfers* shock $\Rightarrow$ income becomes the dominant dimension; bottom 20% *gains* $\sim 10\%$ from transfers

The joint distribution matters — the relevant channel is recession-specific.
Single-margin elasticities mask this.

TO CONCLUDE ...

- ▶ A novel state-space approach for high-frequency, **joint** distributional data — using *any* set of partial/mixed-frequency micro & macro inputs

TO CONCLUDE ...

- ▶ A novel state-space approach for high-frequency, **joint** distributional data — using *any* set of partial/mixed-frequency micro & macro inputs
- ▶ Validated three ways: factor approximation, omitted-wave prediction, and a HANK Monte Carlo

TO CONCLUDE ...

- ▶ A novel state-space approach for high-frequency, **joint** distributional data — using *any* set of partial/mixed-frequency micro & macro inputs
- ▶ Validated three ways: factor approximation, omitted-wave prediction, and a HANK Monte Carlo
- ▶ **Aggregate shocks** drive the bulk of distributional fluctuations ($\geq 70\%$ of every factor)

TO CONCLUDE ...

- ▶ A novel state-space approach for high-frequency, **joint** distributional data — using *any* set of partial/mixed-frequency micro & macro inputs
- ▶ Validated three ways: factor approximation, omitted-wave prediction, and a HANK Monte Carlo
- ▶ **Aggregate shocks** drive the bulk of distributional fluctuations ($\geq 70\%$ of every factor)
- ▶ Recession-time consumption losses are **state-dependent** along the joint $Y \times W$ distribution — neither income nor wealth alone is a sufficient statistic

TO CONCLUDE ...

- ▶ A novel state-space approach for high-frequency, **joint** distributional data — using *any* set of partial/mixed-frequency micro & macro inputs
- ▶ Validated three ways: factor approximation, omitted-wave prediction, and a HANK Monte Carlo
- ▶ **Aggregate shocks** drive the bulk of distributional fluctuations ($\geq 70\%$ of every factor)
- ▶ Recession-time consumption losses are **state-dependent** along the joint $Y \times W$ distribution — neither income nor wealth alone is a sufficient statistic
- ▶ New empirical targets for HANK models; new diagnostic for policy in real time

CONTACT INFORMATION

- ▶ **Reach out!**
- ▶ **Title of the paper:** Distributional Dynamics
- ▶ **Speaker:** Luis Calderon
- ▶ **Email:** luis.calderon@uni-bonn.de
- ▶ **Co-authors:** Christian Bayer and Moritz Kuhn

(2) DEALING WITH PARTIAL OBSERVABILITY OF w

To estimate all $\kappa_{o,t}$, we need to observe all variables $m \in \mathbb{D}$ in dataset j at time t .

What if we only observe $\mathcal{D}^j \subset \mathbb{D}$? Can we still somehow inform the estimation κ_t^j ?

$$dC(u_1, \dots, u_{d-1}) = \int_0^1 dC(u_1, \dots, u_d) du_d = \int_0^1 \sum_{o \in \mathbb{O}} \kappa_o \prod_{m=1}^d Q_{o_m}(u_m) du_d$$

► back

(2) DEALING WITH PARTIAL OBSERVABILITY OF w

To estimate all $\kappa_{o,t}$, we need to observe all variables $m \in \mathbb{D}$ in dataset j at time t .

What if we only observe $\mathcal{D}^j \subset \mathbb{D}$? Can we still somehow inform the estimation κ_t^j ?

$$dC(u_1, \dots, u_{d-1}) = \sum_{o_1} \cdots \sum_{o_{d-1}} \sum_{\mathbf{o}_d} \kappa_{\{o_1, \dots, o_{d-1}, \mathbf{o}_d\}} \left(\prod_{m=1}^{d-1} Q_{o_m}(u_m) \right) \int_0^1 Q_{\mathbf{o}_d}(u_d) du_d$$

► back

(2) DEALING WITH PARTIAL OBSERVABILITY OF w

To estimate all $\kappa_{o,t}$, we need to observe all variables $m \in \mathbb{D}$ in dataset j at time t .

What if we only observe $\mathcal{D}^j \subset \mathbb{D}$? Can we still somehow inform the estimation κ_t^j ?

$$dC(u_1, \dots, u_{d-1}) = \sum_{o_1} \cdots \sum_{o_{d-1}} \kappa_{o_1, \dots, o_{d-1}, 1} \left(\prod_{m=1}^{d-1} Q_{o_m}(u_m) \right)$$

because $\int_0^1 Q_{o_d}(u_d) = \begin{cases} 1 & \text{for } o_d = 1, \\ 0 & \text{otherwise} \end{cases}$

(2) DEALING WITH PARTIAL OBSERVABILITY OF w

To estimate all $\kappa_{o,t}$, we need to observe all variables $m \in \mathbb{D}$ in dataset j at time t .

What if we only observe $\mathcal{D}^j \subset \mathbb{D}$? Can we still somehow inform the estimation κ_t^j ?

$$dC(u_1, \dots, u_{d-1}) = \sum_{o_1} \cdots \sum_{o_{d-1}} \kappa_{o_1, \dots, o_{d-1}, 1} \left(\prod_{m=1}^{d-1} Q_{o_m}(u_m) \right)$$

→ the leading coefficients (in dimension d) and the coefficients from estimating the $d - 1$ dimensional copula are identical.

RECONSTRUCTION

With $\hat{\theta}$, we can generate estimates $\hat{F}_t$ via the **Kalman-filter-smoother**.

With estimates $\hat{F}_t$, we can **reconstruct the data**:

$$\hat{F}_t \xrightarrow[\text{pre-multiply}]{\Gamma} \tilde{\phi}_t \xrightarrow[\text{undo-standardization and add trend}]{\quad} \hat{\hat{\phi}}_t$$

- ▶ This can be done for all t even those without data
- ▶ Posterior draws give us credible bounds on the synthetic data